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Study on Gradation Fractal Cartesian Coordinate System of Asphalt Mixture

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Abstract: In order to explore intrinsic relation of continuous, interlocked-dense and gap gradation, this paper studied them by fractal regression analysis, by which a gradation distribution Cartesian coordinate system was established and slope of the gradation was proposed at the same time. In addition, this study adopted gradation fractal Cartesian coordinate system to analysis the internal relations of continuous, interlocked-dense and gap gradation, at the same time verified and analyzed the correlation between the slope of the gradation and asphalt mixture gradation by using reference experimental data. The results show that: the gradation range given by specification can be approximated by a line segment, and the slope of the line can quantitatively reflect the relative thickness of coarse and fine aggregate of asphalt mixture; the gradation range given by specification possess a small proportion in the gradation fractal Cartesian coordinate system, which can be further optimized, and interlocked-dense asphalt mixture is a transitional form of continuous gradation and gap one; for the SAC asphalt mixture gradation, gradation slope has correlation with the formation of gradation skeleton; the Suitable fractal evaluation model for continuous, gap and interlocked-dense gradation was proposed finally. Thereby, the gradation fractal Cartesian coordinate system can not only reflect the relation of continuous, interlocked-dense and gap gradation, but also make it possible that the gradation distribution range in the specification can be expressed in function.

0 Introductions

Fractal is the essence of aggregate gradation [1]. The fractal of dense-graded aggregate is first-dimension domain, and the fractal of gap-graded aggregate is second-dimension domain. The fractal gradation theory formula was put forward by some scholars, not only used for the continuous gradation aggregate, but also for gap gradation aggregate [2]. For the interlocked-dense gradation of asphalt mixtures, however, the fractal is combination mode between first-dimension domain and second-dimension domain [3]. Fractal evaluation modes of continuous gradation, gap gradation and interlocked-dense gradation asphalt mixtures were proposed by different scholars successively, which provide fractal parameters for quantitative description of aggregation distribution, and establish a relationship between gradation and asphalt mixed material performance. However, the relationship between continuous gradation, gap gradation and interlocked-dense gradation of asphalt mixtures is unclear, and the form of fractal evaluation mode is not uniform. To explore the relationship between the above three gradations of asphalt mixture, their fractal characteristics have been analyzed in this paper.

1 Particle size mass distribution function and fractal mode evaluation of interlocked-dense asphalt mixture

The gradation of interlocked-dense asphalt mixture is combined the mode of single-range fractal and two-range fractals. Particle size-mass distribution function and fractal mode evaluation of interlocked-dense asphalt mixture are as follows [3]:

(1) Particle size-mass distribution function of interlocked-dense asphalt mixture:

$$\mathrm{P}(r)=\left(\frac{r}{r_{\mathrm{NMPS}}}\right)^{3-\mathrm{D_Z}} P_0 \tag{1}$$

$$\mathrm{P}(r)=\left(\frac{r}{r_{NMPS}}\right)^{3-D_C} P_0 \ldots r\in\left[r_{DCF,}r_{NMPS}\right] \tag{2}$$

$$\mathrm{P}(r)=\left(\frac{r}{r_{DCF}}\right)^{3-D_f} P_{DCF} \ldots r\in\left[r_{\min,}r_{DCF}\right] \tag{3}$$

Where, P_r = percentage of passing of r size sieves (%); P_0 = passing percentage of maximum nominal aggregate size sieve (%); r = size sieves of some aggregate (mm); r_{NMPS} = maximum nominal aggregate size (mm); D_z = fractal dimension of gradation; r_{DCF} = particle size of interlocked-dense asphalt mixture gradation at the cut-off point of the coarse and fine aggregate (DCF point) (mm); P_{DCF} = passing percentage at DCF point (%); D_c = fractal dimension of coarse aggregate; D_f = fractal dimension of fine aggregate.

(2) Fractal evaluation mode of interlocked-dense asphalt mixture:

$$\{\mathrm{D_z},\ \mathrm{R_z}^2;\ \mathrm{D_c},\ \mathrm{R_c}^2;\ \mathrm{D_f},\ \mathrm{R_f}^2\} \tag{4}$$

Where D_z, R_z^2=fractal dimension of gradation and correlation coefficient respectively; D_c, R_c^2=fractal dimension of coarse aggregate and correlation coefficients respectively; D_f, R_f^2 = fractal dimension of fine aggregate and correlation coefficients respectively.

2 Establishment of Cartesian coordinate system of asphalt mixture gradation fractal and proposing of gradation slope *k*

In order to comparing the similarities and differences about fractal characteristics of continuous, interlocked-dense and gap gradation, it is found that continuous gradation and gap gradation can be researched in the same way, in which fractal characteristics is used as interlocked-dense gradation. According to 《Chinese highway asphalt pavement construction technical specification》 (JTGF40-2004) [4], AC-16, SAC-16 and SMA-16 were chosen for further study. 5 groups of gradation were chosen from each type of asphalt mixture to make fractal regression calculation: GL1, GQ1 and GJ1; GL2, GQ2 and GJ2; GL3, GQ3 and GJ3; GL4, GQ4 and GJ4; GL5, GQ5 and GJ5 respectively represent the lower limit value, middle value of lower limit and upper limit, middle value, middle value of the middle point value and upper limit value and upper limit value as shown in Table 1. According to Bailey method, the 16mm has been selected as nominal maximum gradation particle size; the boundary of coarse and fine aggregate size (r_{DCF}) of the 15 groups of gradation is 4.75mm [5].

Table 1 15 groups of AC-16, SAC-16 and SMA-16 mixture gradation

Gradation type		Sieve size [mm]										
		19	16	13.2	9.5	4.75	2.36	1.18	0.6	0.3	0.15	0.075
AC -16	GL1	100	90	70	60	34	20	13	9	7	5	4
	GL2	100	92.5	75.5	65	41	27	18.75	13.25	9.75	7.25	5
	GL3	100	95	81	70	48	34	24.5	17.5	12.5	9.5	6
	GL4	100	97.5	86.5	75	55	41	30.25	21.75	15.25	11.75	7
	GL5	100	100	92	80	62	48	36	26	18	14	8
SAC -16	GQ1	100	95	79	58	30	23	17	13	10	8	6
	GQ2	100	96.25	80.75	60.25	32.5	25.25	19	14.75	11.5	9.25	7
	GQ3	100	97.5	82.5	62.5	35	27.5	21	16.5	13	10.5	8
	GQ4	100	98.75	84.25	64.75	37.5	29.75	23	18.25	14.5	11.75	9
	GQ5	100	100	86	67	40	32	25	20	16	13	10
SMA -16	GJ1	100	90	65	45	20	15	14	12	10	9	8
	GJ2	100	92.5	70	50	23	17.25	16	13.5	11.25	10.25	9
	GJ3	100	95	75	55	26	19.5	18	15	12.5	11.5	10
	GJ4	100	97.5	80	60	29	21.75	20	16.5	13.75	12.75	11
	GJ5	100	100	85	65	32	24	22	18	15	14	12

The data in Table 1 is calculated by fractal regression, and a parameter k is introduced, where $k = D_c / D_f$. Results are shown in Table 2.

Table 2 shows that Rz^2, namely correlation coefficients of AC-16, SAC-16 and SMA-16 gradually decreases, but the correlation coefficients R_z^2 of AC-16 and SAC-16 are with little difference, and that of SMA-16 is relatively lower; At the same time, the gradation correlation coefficients of coarse and fine aggregate (R_c^2, R_f^2) of AC-16, SAC-16 and SMA-16 are all greater than 0.978. The result indicates: Firstly, the continuous gradation is first-dimension domain fractal mode, and the gradation of interlocked-dense asphalt mixture is combination mode of first-dimension domain fractal and second-dimension domain fractal. Secondly, AC-16, SAC-16 and SMA-16 are all with the characteristics of second-dimension domain fractal. The D_c and D_f of AC-16 and SAC-16 vary in the same trend, namely the value of D_c and D_f is increase with the group number up; but the D_c and D_f of SMA-16 change in the opposite trend. However k values of AC-16, SAC-16 and SMA-16 asphalt mixture gradation vary in the same trend which value goes up with the increase of the group number. Meanwhile, k value of AC-16 is max one and most close to 1, and k value of SMA-16 is min one, but k value of SAC-16 are in the middle of the two. That means: to AC-16, D_c and D_f are very close, not necessarily equal, and can be considered in accordance with second-dimension domain fractal; AC-16, SAC-16 and SMA-16 may be linked by k value. Further analysis on AC, SAC and SMA with different nominal maximum particle size can get the consistent law. Therefore, continuous, interlocked-dense and gap gradation are all with the characteristics of second-dimension domain fractal be considered in accordance with second-dimension domain fractal.

Table 2 Results of fractal regression calculation for 15 groups of AC-16, SAC-16 and SMA-16 mixture gradation

Gradation type		k_f	D_f	R_f^2	k_c	D_c	R_c^2	k_z	D_z	R_z^2	k
AC-16	GL1	0.507	2.493	0.981	0.772	2.228	0.990	0.600	2.400	0.982	0.89
	GL2	0.496	2.504	0.997	0.644	2.356	0.993	0.541	2.459	0.995	0.94
	GL3	0.489	2.511	0.997	0.537	2.463	0.995	0.499	2.501	0.998	0.98
	GL4	0.484	2.516	0.993	0.455	2.545	0.994	0.466	2.534	0.996	1.01
	GL5	0.481	2.519	0.988	0.365	2.635	0.986	0.439	2.561	0.990	1.05
SAC-16	GQ1	0.386	2.614	0.998	0.899	2.101	0.992	0.511	2.489	0.967	0.80
	GQ2	0.368	2.632	0.999	0.843	2.157	0.991	0.483	2.517	0.969	0.82
	GQ3	0.353	2.647	0.999	0.791	2.209	0.989	0.459	2.541	0.971	0.83
	GQ4	0.331	2.669	0.999	0.698	2.302	0.984	0.420	2.580	0.975	0.86
	GQ5	0.341	2.659	0.999	0.743	2.257	0.987	0.438	2.562	0.973	0.85
SMA-16	GJ1	0.212	2.788	0.980	1.186	1.814	0.996	0.449	2.551	0.861	0.65
	GJ2	0.217	2.783	0.980	1.088	1.912	0.995	0.434	2.566	0.899	0.69
	GJ3	0.221	2.779	0.979	1.003	1.997	0.992	0.421	2.579	0.888	0.72
	GJ4	0.225	2.775	0.979	0.861	2.139	0.976	0.409	2.591	0.876	0.77
	GJ5	0.227	2.773	0.978	0.861	2.139	0.976	0.399	2.601	0.908	0.77

2.1 Establishment of Cartesian coordinate system of asphalt mixture gradation fractal

In view of the second-dimension domain fractal characteristic of continuous gradation, interlocked-dense and gap gradation asphalt mixture, it can be considered to analyze through second-dimension domain fractal. A point G of (D_f, D_c) can be used to indicate a gradation in the coordinate system as shown in Fig.1, which D_f is as the abscissa axis and D_c as the longitudinal axis. That is the gradation fractal Cartesian coordinate system of asphalt mixture. 15 groups of coarse and fine aggregate fractal dimension in Table 2 can be shown with points in the coordinate system (D_f, D_c) in Fig.2. The type gradation of asphalt mixture gradation scopes for AC-16, SAC-16 and SMA-16 can be approximately expressed as a line segment, so it is more conducive to compare and analyze the relationship between asphalt mixture gradations.

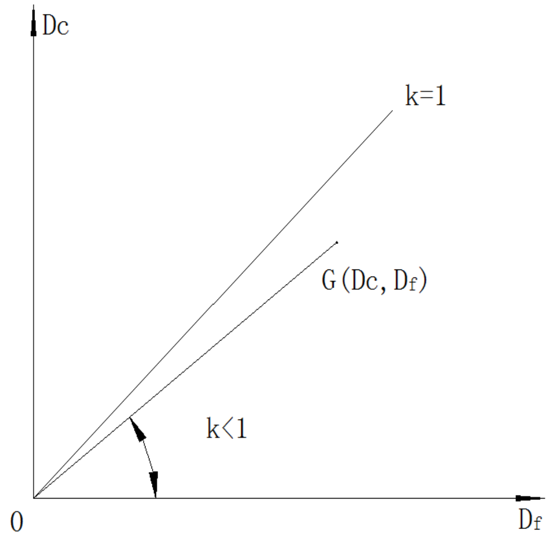

Fig. 1 Gradation fractal Cartesian coordinate system of asphalt mixture

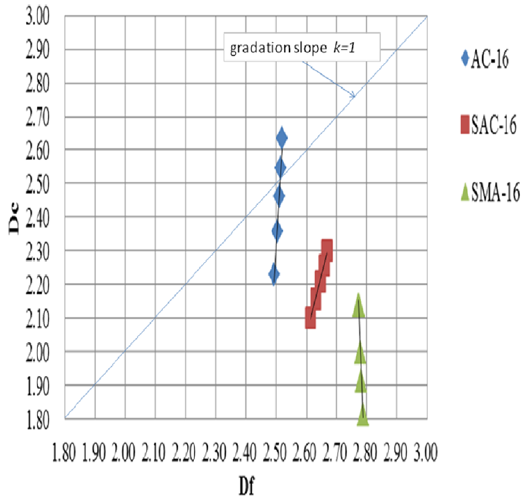

Fig. 2 Position of gradations of Table 1 in Gradation fractal Cartesian coordinate system

2.2 Gradation slope *k*.

Connect point O and G in Fig.1, then the slope of the line OG is the parameter k, where $k = D_c/D_f$. Therefore, the geometrical meaning of the parameter k is the slope of connection line of one gradation point and the origin of the coordinate. Analysis of variation of the k values in Table 2 shows that the actual meaning of the parameter k reflects the relative degree and the proportion of coarse and fine aggregate ($k \leq 1$).

3 Application of gradation fractal Cartesian coordinate system of asphalt mixture

3.1 Comparative analysis of different nominal maximum particle size of the asphalt mixture gradation

In order to analyze the relationship between continuous gradation, interlocked-dense asphalt mixture gradation and gap gradation with different nominal maximum aggregate size, Cartesian coordinate system of asphalt mixture gradation fractal is used. According to 《Technical Specifications for Construction of Highway Asphalt Pavements》 (JTGF40-2004) [4], AC-13, SAC-13, SMA-13, AC-20, SAC-20, SMA-20 asphalt mixture gradations are selected to do fractal regression calculation. The lower, the median and the upper limits of AC-13, SAC-13 and SMA-13-type asphalt mixture gradation are GL9, GQ9 GJ9, GL10, GQ10 and GJ10, GL11, GQ 11 and GJ11, which are shown in Tables3 and 4 respectively. In accordance with the provisions of the limit size of coarse and fine aggregate from Bailey method [5], when nominal maximum aggregates size are 13mm and 19mm, r_{DCF} are 2.36mm and 4.75mm respectively.

Table 3 9 groups of AC-13, SAC-13 and SMA-13 asphalt mixture aggregate gradation

Gradation type		Sieve size [mm]									
		16	13.2	9.5	4.75	2.36	1.18	0.6	0.3	0.15	0.075
AC-13	GL6	100	90	68	38	24	15	10	7	5	4
	GL7	100	95	76.5	53	37	26.5	19	13.5	10	6
	GL8	100	100	85	68	50	38	28	20	15	8

SAC-13	GQ6	100	95	60	30	22	16	12	10	8	6
	GQ7	100	97.5	70	35	26.5	20	16	14	12	8
	GQ8	100	100	80	40	31	24	20	18	16	10
SMA-13	GJ6	100	90	50	20	15	14	12	10	9	8
	GJ7	100	95	62.5	27	20.5	19	16	13	12	10
	GJ8	100	100	75	34	26	24	20	16	15	12

Table 4 9 groups of AC-20, SAC-19 and SMA-19 asphalt mixture aggregate gradation

Gradation type		Sieve size [mm]											
		26.5	19	16	13.2	9.5	4.75	2.36	1.18	0.6	0.3	0.15	0.075
AC -20	GL9	100	90	74	62	50	26	16	12	8	5	4	3
	GL10	100	95	83	72	61	41	30	22.5	16	11	8.5	5
	GL11	100	100	92	82	72	56	44	33	24	17	13	7
SAC -20	GQ9	100	95	80	65	50	30	24	17	12	10	8	5
	GQ10	100	97.5	87.5	72.5	57.5	35	29.5	20.5	16	13.5	11.5	7.5
	GQ11	100	100	95	80	65	40	35	24	20	17	15	10
SMA -20	GJ9	100	90	72	62	40	18	13	12	10	9	8	8
	GJ10	100	95	82	72	47.5	24	17.5	16	13	11.5	10.5	10
	GJ11	100	100	92	82	55	30	22	20	16	14	13	12

After regression calculation to fractal characteristic for the gradations Table 3 and 4, the results of Table 3and 4 are shown in Table 5 and Table 6 respectively.

Table 5 Fractal characteristic for the gradations of Table 3

Gradation type	AC-13			SAC-13			SMA-13		
	GL6	GL7	GL8	GQ6	GQ7	GQ8	GJ6	GJ7	GJ8
D_c	2.235	2.469	2.629	2.152	2.244	2.321	1.934	2.097	2.217
R_c^2	0.989	0.997	0.997	0.968	0.97	0.965	0.953	0.962	0.962
D_f	2.478	2.488	2.491	2.637	2.683	2.711	2.808	2.786	2.733
R_f^2	0.987	0.993	0.977	0.993	0.972	0.941	0.988	0.985	0.981
D_z	2.378	2.488	2.523	2.483	2.545	2.588	2.55	2.582	2.606
R_z^2	0.985	0.998	0.987	0.948	0.945	0.943	0.829	0.872	0.902
k	0.9	0.99	1.06	0.82	0.84	0.86	0.69	0.75	0.8

Table 6 Fractal characteristic for the gradations of Table 4

Gradation type	AC-20			SAC-19			SMA-19		
	GL9	GL10	GL11	GQ9	GQ10	GQ11	GJ9	GJ10	GJ11
D_c	2.192	2.454	2.632	2.251	2.344	2.424	1.952	2.114	2.238
R_c^2	0.984	0.982	0.963	0.977	0.965	0.940	0.976	0.959	0.929
D_f	2.477	2.506	2.517	2.581	2.642	2.680	2.809	2.795	2.785
R_f^2	0.991	0.993	0.981	0.989	0.984	0.977	0.926	0.944	0.952
D_z	2.375	2.499	2.566	2.494	2.556	2.599	2.538	2.571	2.596
R_z^2	0.986	0.997	0.986	0.982	0.978	0.976	0.863	0.889	0.906
k	0.88	0.98	1.05	0.87	0.89	0.9	0.69	0.76	0.8

According to the fractal regressive computation based on Table 2, Table 5 and Table 6, the values can be expressed by grading coordinates in the Cartesian coordinate system of the asphalt mixture gradation that are lower limit, mid-value and upper limit of gradation in Table 1 as well as gradation listed in Table 3 and 4 as shown in Fig. 3. It is indicated that the grading distribution range of the nine groups of gradation could be described approximately in 9 straight lines, and lines intersect that represent the grading distribution ranges for the asphalt mixture SMA-13 and SMA-16. Meantime, the corresponding lines for grading distribution range of the same gradation type of the asphalt mixture with different nominal maximum aggregate size are placed densely. The above-mentioned implies that the rational distribution with good optimizing potential range given in the specification occupies a small portion in the Cartesian coordinate system of the asphalt mixture gradation; and the same type of the asphalt mixture gradation with different NMAS may have the same fractal characters. It is implied from the distribution of the continuous, interlocked-dense and gap grading asphalt mixture that the grading distribution area of the interlocked-dense asphalt mixture is between those continuous and gap grading and the three types have a common value area of D_c but no for D_f. This can be explained that the interlocked-dense type is the transitional form between the continuous grading and the gap grading one; the fine aggregate exert greater effect than the coarse aggregate on the grading type of asphalt mixture. It can be seen in the analysis of the grading slope represented as k that gap grading has smaller value than that of the continuous grading while the interlocked-dense falls in between; those three with different NMAS share almost the same value area of k. The above shows that the matching of the fine and coarse aggregate for the interlocked-dense grading of the asphalt mixture has something in common with the continuous grading and gap grading ones which happened to hold the same view that the SAC mineral aggregate grading could be designed as interlocked-dense or the common framework-dense type and even dense-suspended structure [6]. As for the continuous grading, interlocked-dense grading and gap grading with different NMAS, they stay almost the same respectively in their relative degree of thickness of the aggregate.

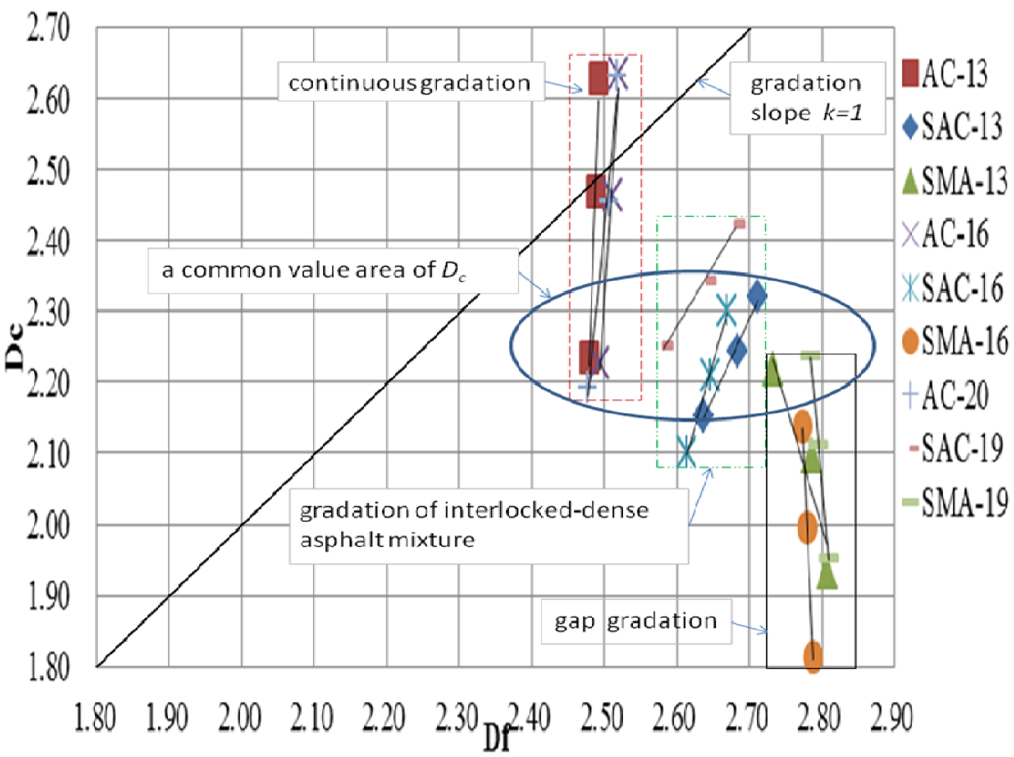

Fig. 3 Position of gradations of Table 1, 3 and 4 in Gradation fractal Cartesian coordinate system

3.2 Mathematics express of gradation range

The gradation distribution range in the specification can be expressed in function: $y=kx+c$, since it could be described approximately as straight lines and the grading type depends majorly on the fractal dimension of the fine aggregate. The 9 function forms for the gradation distribution in Fig. 3 are shown in Table 7.

Table 7 Functions of gradation range

Gradation type	Function	R^2	Range of x
SAC-16	y = 3.624x - 7.376	0.994	[2.614,2.669]
SAC-19	y = 1.726x - 2.208	0.991	[2.581,2.680]
SMA-13	y = -3.495x + 11.78	0.900	[2.733,2.808]
SMA-16	y = -21.56x + 61.92	0.998	[2.773,2.788]
SMA-19	y = -11.89x + 35.36	0.999	[2.785,2.809]
AC-13	y = 28.56x - 68.56	0.963	[2.478,2.491]
AC-16	y = 10.6x - 24.07	0.979	[2.493,2.519]
AC-20	y = 15.21x - 35.71	0.983	[2.477,2.517]
SAC-13	y = 2.255x - 3.799	0.992	[2.637,2.711]

3.3 Correlation analysis of grading slope *k* and asphalt mixture grading

To verify the correlation of grading slope k and asphalt mixture gradation, refer to the test data from reference [7], as shown in Table 8.

Table 8 Reference data

Gradation type	D_c	D_f	V_{co}	V_f	k	V_f-V_{co}
A	2.2995	2.5140	0.087	0.075	0.91	-0.012
B	2.3897	2.4672	0.091	0.110	0.97	0.019
C	2.4675	2.4455	0.092	0.151	1.01	0.059

In Table 8, V_{co} is the porosity of coarse aggregate fractal; V_f is the fractal volume of fine aggregate in coarse aggregate; k is the gradation slope. It can be seen from the Table 8: Change trend of k and V_f-V_{co} are consistent from gradation A to C, and V_f-V_{co} >0 when k increases to a certain value, so the asphalt mixture could not form the skeleton type. By contrasting the gradation slope k of SAC-13, SAC-16 and SAC-19, it can be drawn a conclusion that $0.8 \le k \le 0.9$, and the smaller the k, the closer the grading of SAC and SMA. Grading A is minimum and the closest to the 3 gradations of SAC, so the grading is reasonable and form the skeleton under the condition of optimum asphalt content. The grading of B and C is close to continuous grading and could not form skeleton owing to the larger k value. For SAC, k value is smaller, the gradation of SAC is closer to SMA, and coarse aggregate is easy to form the skeleton. Conversely, k value is bigger, the gradation of SAC is closer to AC, and coarse aggregate is hard to form skeleton.

4. The optimization of asphalt mixture gradation fractal evaluation model

Gradation slope can quantitatively reflect the thickness ratio of asphalt mixture gradation on the basis of second-dimension domain fractal features of continuous gradation, interlocked-dense gradation and the gap grading. It proposed that the fractal evaluation model is appropriate for continuous gradation, interlocked-dense gradation and the gap gradation combined with existing gradation fractal evaluation model, as follows:

$$\{ D_c, R_c^2; D_f, R_f^2; k \} \tag{5}$$

Where, D_c, R_c^2 = fractal dimension and correlation coefficient of coarse aggregate respectively; D_f, R_f^2= fractal dimension and correlation coefficient of fine aggregate respectively; k = gradation slope, $k=D_c/D_f$.

5. Conclusions

From the fractal property analysis of the continuous gradation, interlocked dense asphalt mixture gradation and gap gradation, the following conclusions can be reached:

(1) Continuous gradation, interlocked-dense asphalt mixture gradation and gap gradation can be considered by second-dimension domain fractal. On this basis, this paper established the asphalt mixture gradation fractal Cartesian coordinate system, and put forward the concept of gradation slope. Asphalt mixture gradation can be expressed as gradation coordinate points (D_c, D_f), and gradation scope can be approximately expressed as a line segment. Gradation slope k describes relative thickness of coarse aggregate and fine aggregate of asphalt mixture gradation. In general, $k \le 1$, the minimum value of k and the correlation between k and asphalt mixture need to further research.

(2) The gradation scope supplied by specifications is very limited in the grading system, and asphalt mixture gradation has a plenty of space for optimization. Interlocked-dense asphalt mixture gradation is a kind of transition from continuous gradation to gap gradation. Fine aggregate has a more significant effect on asphalt mixture gradation type than coarse one.

(3) For continuous gradation, interlocked-dense asphalt mixture gradation and gap gradation with different nominal maximum aggregate size, the relative thickness of coarse aggregate and fine aggregate is essentially identical.

(4) For SAC asphalt mixture gradation, as the values of D_c and D_f are in a reasonable range, with the decreasing of gradation slope k, SAC asphalt mixture gradation approach SMA asphalt mixture gradation, and coarse aggregate is easy to form skeleton. With the increasing of gradation slope k, SAC asphalt mixture gradation approach AC asphalt mixture gradation, and coarse aggregate is not easy to form skeleton.

(5) This paper put forward an evaluation model is suitable for continuous gradation, interlocked dense asphalt mixture gradation and gap gradation.

Acknowledgments

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